WEEK 3 PROJECT

REGRESSION ANALYSIS

Build and interpret a regression model
Explain coefficients in plain language

Project detail	Description
Project title	House Price Prediction Using Linear Regression
Problem type	Supervised Machine Learning – Regression
Target	House price (■)
Input features	Area, bedrooms, age and distance to city
Model	Multiple Linear Regression
Tools	Python, Pandas, NumPy, Matplotlib, Scikit-learn
Dataset	300 simulated sample records

Complete project report with concepts, workflow, Python code, model evaluation, coefficient interpretation, graphs, conclusion, testing checklist and viva questions.

1. Project Overview

Task requirement: Build and interpret a regression model; explain coefficients in plain language.

Regression is a supervised machine-learning technique used to predict a continuous numerical value. Examples include house prices, sales, salary and delivery time.

Project scenario: A real-estate company wants to estimate house prices using property information. This project uses area, number of bedrooms, house age and distance from the city to predict price.

Aim: Train a regression model, predict prices for unseen test records, evaluate errors and explain the model coefficients.

The dataset is simulated for educational demonstration. It is not real company data and should not be used for actual property valuation.

Learning objectives

• Understand regression and supervised learning.

• Identify independent variables (features) and dependent variable (target).

• Prepare data and split it into training and testing sets.

• Train a Multiple Linear Regression model.

• Evaluate predictions using MAE, RMSE and R^2.

• Explain coefficients and intercept in simple language.

• Visualise model performance.

2. Regression Concepts

2.1 Definition

Regression is a statistical and machine-learning method that learns the relationship between input variables and a continuous output. It predicts a number rather than a category.

2.2 Types of regression

Type	Explanation	Example
Simple Linear	One input feature predicts a number.	Price from area.
Multiple Linear	Two or more input features predict a number.	Price from area, bedrooms, age and distance
Polynomial	Uses powers of features to model curves.	Non-linear growth.
Ridge	Linear regression with L2 regularisation.	Reduce large coefficients.
Lasso	Linear regression with L1 regularisation.	Can shrink some coefficients to zero.

2.3 Regression vs Classification

Regression	Classification
Predicts continuous numerical values.	Predicts categories or labels.
Example: house price ■45 lakh.	Example: email is spam or not spam.
Metrics: MAE, RMSE, R^2.	Metrics: accuracy, precision, recall, F1-score.

3. Problem Definition

Business question: Can house prices be estimated from basic property characteristics?

Target variable (y): House_Price.

Input features (X): Area_sqft, Bedrooms, Age_years and Distance_to_city_km.

• Create a structured dataset with property details and prices.

• Inspect the data and select input and target columns.

• Split the data into training and testing sets.

• Train a Multiple Linear Regression model.

• Predict prices for the test set.

• Evaluate prediction errors and interpret coefficients.

4. Dataset Description

The project uses 300 simulated house records. Each row represents one house. The target price is generated from an example relationship plus random noise.

Column	Description	Role
Area_sqft	House area in square feet.	Input
Bedrooms	Number of bedrooms.	Input
Age_years	Age of the house in years.	Input
Distance_to_city_km	Distance from city centre in kilometres.	Input
House_Price	House price in Indian rupees.	Target

Area	Bedrooms	Age	Distance	Price (■)
769	2	25	2.0	185,164
2071	5	31	22.2	441,855
1844	1	30	18.0	298,822
1434	5	5	5.2	328,948
1423	3	33	3.2	291,026

A fixed random seed is used so that the same code can reproduce the same sample data and results.

5. Tools and Technologies

Tool	Purpose
Python	Main programming language.
Pandas	Create, inspect and manipulate tabular data.
NumPy	Numerical calculations and sample data generation.
Matplotlib	Create graphs and visualisations.
Scikit-learn	Split data, train regression model and evaluate it.
Google Colab / Jupyter	Run the Python notebook.

6. Project Workflow

1. Define the prediction problem

2. Create or load the dataset

3. Inspect and prepare the data

4. Select features (X) and target (y)

5. Split into training and testing sets

6. Train the Linear Regression model

7. Predict prices on test data

8. Evaluate MAE, RMSE and R^2

9. Interpret coefficients and visualise results

10. Write conclusion and limitations

Workflow: Data → Preparation → Train/Test Split → Model Training → Prediction → Evaluation → Interpretation → Conclusion

7. Data Preparation

Data preparation checks that the dataset is suitable for model training. In this simulated dataset, all rows are generated with complete values. For real data, inspect missing values, duplicates, incorrect data types and unusual values.

• df.head() displays the first records.

• df.info() checks column names and data types.

• df.isnull().sum() counts missing values.

• df.duplicated().sum() checks duplicate rows.

• Separate features from the target variable.

Do not include the target column in the input features. Doing so would leak the answer to the model and produce misleading evaluation results.

8. Train-Test Split

Set	Percentage	Rows	Purpose
Training	80%	240	Used to learn the relationship.
Testing	20%	60	Used to evaluate unseen records.

The project uses train_test_split(test_size=0.2, random_state=42). The random state makes the split reproducible. The test set is not used to train the model.

9. Multiple Linear Regression

Multiple Linear Regression predicts a continuous target using several input features.

General equation: Y = β■ + β■X■ + β■X■ + β■X■ + β■X■ + ε

Y = house price; β■ = intercept; β values = coefficients; X values = area, bedrooms, age and distance; ε = unexplained variation.

Fitted model equation from this run:
Price = 48,838 + (180.62 × Area) + (12,138.89 × Bedrooms) + (-883.61 × Age) + (-1,569.07 × Distance)

The fitted coefficients can differ from the data-generation values because the model learns from a training sample that includes random noise.

10. Complete Python Implementation

```
# WEEK 3: REGRESSION PROJECT
# House Price Prediction using Multiple Linear Regression

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error, r2_score

# 1. Generate a reproducible sample dataset
rng = np.random.default_rng(42)
n = 300
area = rng.integers(600, 2501, n)
bedrooms = rng.integers(1, 6, n)
age = rng.integers(0, 41, n)
distance = rng.uniform(1, 25, n)
noise = rng.normal(0, 18000, n)
price = (50000 + 180*area + 12000*bedrooms
         - 900*age - 1500*distance + noise)

df = pd.DataFrame({
```

```
    "Area_sqft": area,
    "Bedrooms": bedrooms,
    "Age_years": age,
    "Distance_to_city_km": distance.round(1),
    "House_Price": price.round(0).astype(int)
})

# 2. Inspect the data
print(df.head())
print(df.info())
print(df.isnull().sum())

# 3. Select features and target
X = df[["Area_sqft", "Bedrooms",
        "Age_years", "Distance_to_city_km"]]
y = df["House_Price"]
```

10. Python Implementation (continued)

```
# 4. Split: 80% train and 20% test
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 5. Train the model
model = LinearRegression()
model.fit(X_train, y_train)

# 6. Predict on unseen test records
y_pred = model.predict(X_test)

# 7. Evaluate the model
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)
print("MAE:", round(mae, 2))
print("RMSE:", round(rmse, 2))
print("R-squared:", round(r2, 4))

# 8. Display intercept and coefficients
print("Intercept:", model.intercept_)
for feature, value in zip(X.columns, model.coef_):
    print(feature, "coefficient:", value)

# 9. Actual vs predicted graph
plt.scatter(y_test, y_pred, alpha=0.7)
low = min(y_test.min(), y_pred.min())
high = max(y_test.max(), y_pred.max())
plt.plot([low, high], [low, high], "--")
plt.xlabel("Actual price (Rs.)")
plt.ylabel("Predicted price (Rs.)")
plt.title("Actual vs Predicted House Prices")
plt.grid(alpha=0.3)
plt.show()

# 10. Residual plot
residuals = y_test - y_pred
plt.scatter(y_pred, residuals, alpha=0.7)
plt.axhline(0, linestyle="--")
plt.xlabel("Predicted price (Rs.)")
plt.ylabel("Residual")
plt.title("Residual Plot")
plt.grid(alpha=0.3)
plt.show()

# 11. Predict a new example house
new_house = pd.DataFrame({
    "Area_sqft": [1500],
    "Bedrooms": [3],
    "Age_years": [5],
    "Distance_to_city_km": [8.0]
})
print("Predicted price:", model.predict(new_house)[0])
```

11. Model Evaluation Metrics

Metric	Meaning	Interpretation
MAE	Average absolute difference between actual and predicted values.	Lower is better; measured in rupees.
RMSE	Square root of average squared prediction errors.	Lower is better; penalises large errors.
R^2	Proportion of target variation explained by the model.	Closer to 1 is generally better; can be negative.

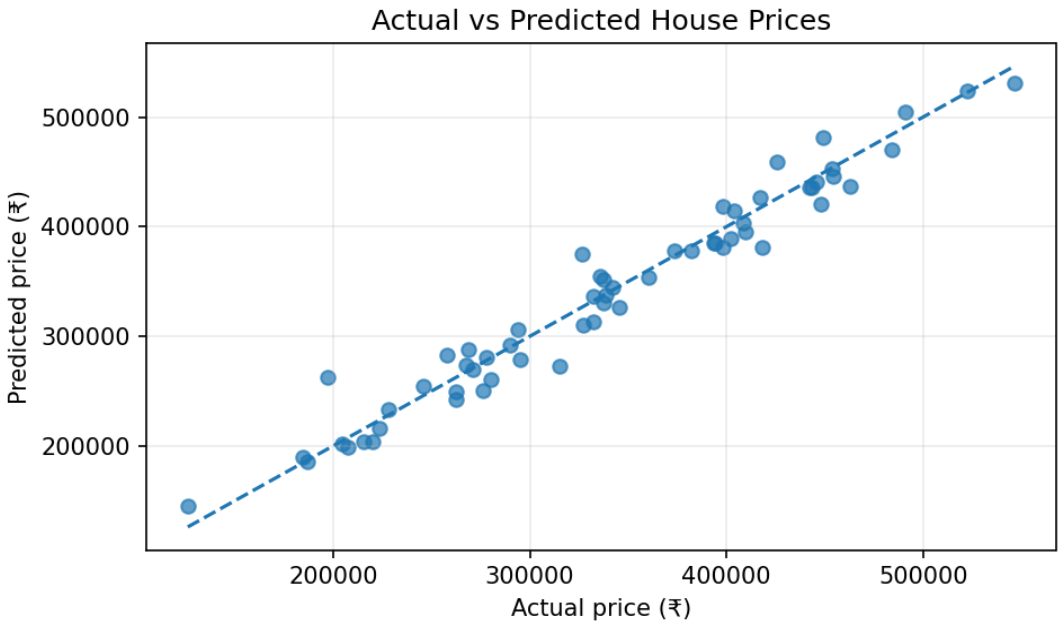
Formulas:
MAE = (1/n) Σ |Actual – Predicted|
RMSE = √[(1/n) Σ (Actual – Predicted)²]
$R^2 = 1 – [\Sigma(\text{Actual} – \text{Predicted})^2 / \Sigma(\text{Actual} – \text{Mean Actual})^2]$

12. Actual Results from This Run

Evaluation item	Result
Total records	300
Training records	240
Testing records	60
MAE	■14,331.84

Evaluation item	Result
RMSE	■18,857.78
R-squared	0.9602

The test R^2 is 0.9602. In this example, the model explains approximately 96.0% of the variation in test-set house prices. MAE gives the average absolute prediction error in rupees; RMSE gives more weight to large errors.



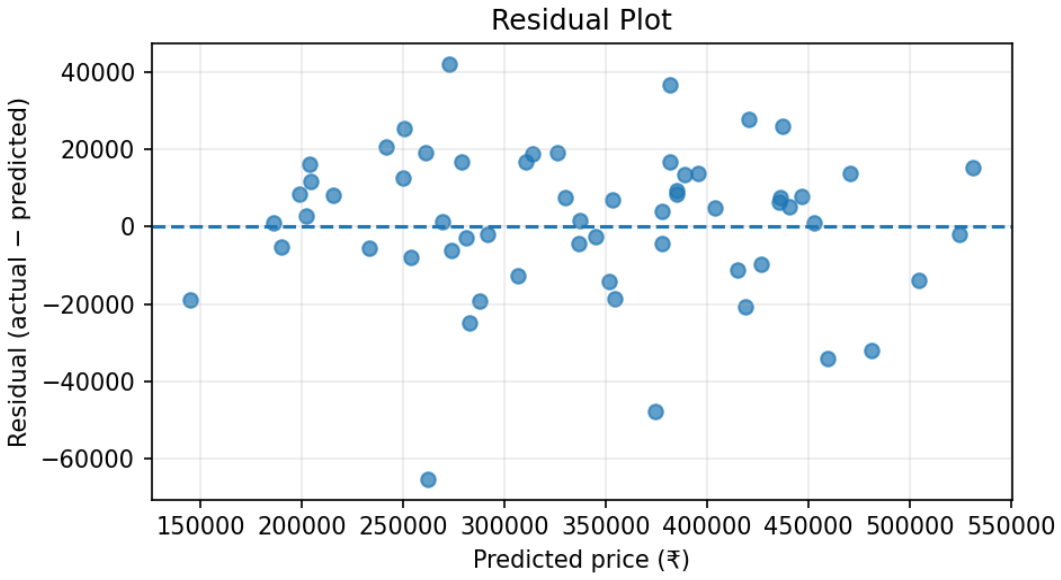
In the actual-versus-predicted plot, points closer to the diagonal line indicate more accurate predictions.

13. Residual Analysis

A residual is the difference between an actual value and its predicted value.

Residual = Actual value – **Predicted value.**

A random cloud around zero is generally desirable. Curved patterns may suggest non-linearity; a widening spread may suggest unequal error variance; distant points may indicate outliers.

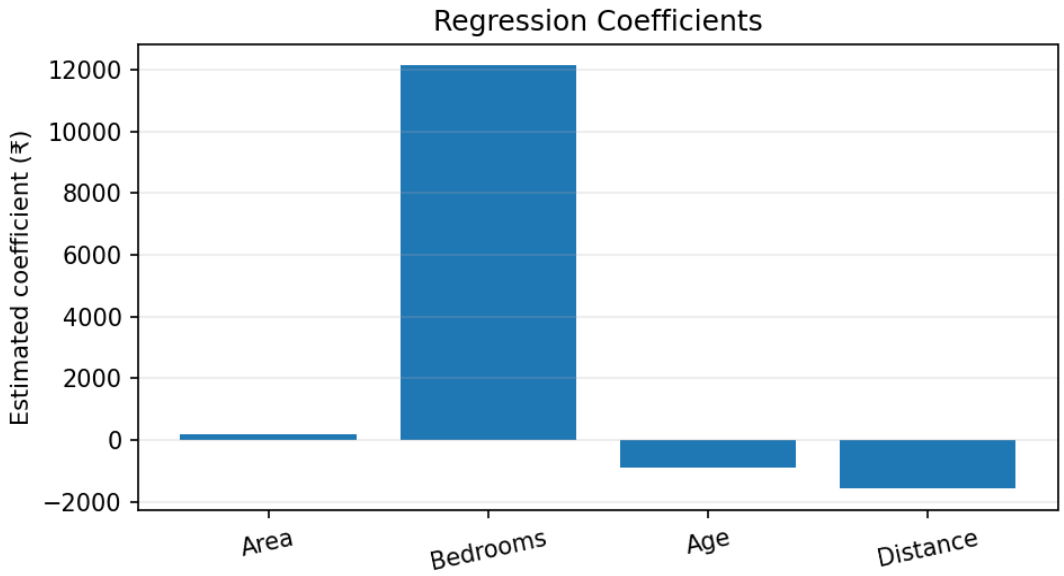


14. Coefficient Interpretation in Plain Language

A regression coefficient estimates how much the predicted house price changes when one feature increases by one unit, while the other included features are held constant.

Feature	Coefficient (■)	Plain-language meaning
Area_sqft	180.62	Change in predicted price for one extra square foot, holding other feat
Bedrooms	12,138.89	Change in predicted price for one extra bedroom, holding other featur
Age_years	-883.61	Change in predicted price for one additional year of house age, holdin
Distance_to_city_km	-1,569.07	Change in predicted price for one additional kilometre from the city, ho

A positive coefficient means the predicted price increases as that feature increases, with other features fixed. A negative coefficient means the predicted price decreases. These are modelled associations, not proof of cause and effect.



14.1 Intercept

The intercept is approximately ■48,838.24. It is the predicted price when all input features are zero. A zero-area house is not realistic, so the intercept is mainly a mathematical starting point.

15. Example Prediction

Input	Example value
Area	1,500 square feet
Bedrooms	3
Age	5 years
Distance to city	8 km
Predicted price	■339,221.66

This is a model prediction for an example record, not a professional property valuation. Real prices depend on location, condition, amenities, demand and other factors.

16. Advantages of Regression

• Predicts continuous numerical values.

• Helps identify relationships between features and an outcome.

• Linear regression coefficients are relatively easy to explain.

• Supports planning, forecasting and data-driven decisions.

• Provides a useful baseline for more advanced models.

17. Limitations and Assumptions

• The simulated dataset does not represent every real-world housing factor.

• Linear regression assumes an approximately linear relationship.

• Outliers can influence the fitted model.

• Strong multicollinearity can make coefficient estimates unstable.

• Errors should be independent and have roughly constant variance for standard inference.

• Prediction quality may fall in a different market or time period.

• R^2 alone does not prove the model is suitable; inspect errors and residuals too.

18. Applications of Regression

• House price prediction.

• Sales and revenue forecasting.

• Demand and inventory planning.

• Salary and cost estimation.

• Weather and energy consumption forecasting.

• Healthcare measurement prediction.

• Business trend analysis.

19. Testing Checklist

• Dataset contains 300 rows and the expected columns.

• Input features do not include the target column.

• Data is split into 80% training and 20% testing.

• Model is trained only on training data.

• Predictions are generated for test records.

• MAE, RMSE and R^2 are printed.

• Model coefficients and intercept are displayed.

• Actual-versus-predicted and residual graphs are generated.

• New-house prediction code runs without errors.

• Conclusion explains results and limitations in plain language.

20. Google Colab Execution Steps

• Open https://colab.research.google.com/ and choose New Notebook.

• Rename it Week3_Regression_Project.

• Paste the complete Python code from Section 10 into a code cell.

• Run the cell and wait for the dataset, metrics and graphs.

• Check coefficient values and the example prediction.

• Save the notebook to Google Drive.

• Download the notebook as .ipynb if requested.

• Attach screenshots of code output and graphs with this report.

21. Short Submission Description

I completed the Week 3 Regression project by building a Multiple Linear Regression model to predict house prices using area, number of bedrooms, house age and distance from the city. I generated and inspected a sample dataset, divided it into training and testing sets, trained the model using Scikit-learn, and evaluated it using MAE, RMSE and R-squared. I also plotted actual versus predicted values, analysed residuals and interpreted regression coefficients in plain language. This project helped me understand how regression models make numerical predictions and how to evaluate their performance.

22. Viva Questions and Answers

1. What is regression?
A supervised learning method used to predict continuous numerical values.

2. What is a regression model?
A mathematical model that estimates the relationship between input features and a numerical target.

3. What is linear regression?
A method that fits a linear relationship between features and a continuous target.

4. What is multiple linear regression?
Linear regression that uses two or more input features.

5. What is the target in this project?
House_Price.

6. What are the input features?
Area, bedrooms, age and distance to the city.

7. What is training data?
Data used by the model to learn patterns and estimate coefficients.

8. What is testing data?
Held-out data used to evaluate the trained model.

9. Why split the dataset?
To evaluate how well the model predicts records it did not train on.

10. What is a coefficient?
The estimated change in prediction for a one-unit increase in a feature, holding other features fixed.

11. What is an intercept?
The predicted target when all input features are zero.

12. What is MAE?
Mean Absolute Error, the average absolute prediction error.

13. What is RMSE?
Root Mean Squared Error, which penalises large prediction errors more strongly.

14. What is R-squared?
A measure of target variation explained by the model on the evaluated data.

15. What is a residual?
Actual value minus predicted value.

16. Which library trains the model?
Scikit-learn.

17. Why use Pandas?
To create, inspect and manipulate tabular data.

18. Why use Matplotlib?
To create graphs and visualise model performance.

19. What does a negative coefficient mean?
The predicted target decreases as that feature increases, with other included features held constant.

20. Does regression prove causation?
No. A fitted relationship alone does not prove that one variable causes another.

23. Conclusion

A Multiple Linear Regression model was developed to predict house prices from four features: area, bedrooms, age and distance to the city. The data was split into 240 training records and 60 testing records. The model achieved an R^2 of 0.9602, MAE of ■14,331.84 and RMSE of ■18,857.78 on the test set. Coefficients were interpreted to explain how each input is associated with the predicted price when other features are held constant. The project demonstrates the complete regression workflow, from data preparation to evaluation and interpretation.

The results are from an illustrative simulated dataset. A real-world model should be trained and validated using reliable housing data and additional relevant features.

24. References

• Scikit-learn Linear Regression: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

• Scikit-learn model evaluation: https://scikit-learn.org/stable/modules/model_evaluation.html

• Pandas documentation: https://pandas.pydata.org/docs/

• NumPy documentation: https://numpy.org/doc/

• Matplotlib documentation: https://matplotlib.org/stable/